Aniket Das

PhD Student in Computer Science, Stanford University

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Research Interests

Markov Chains, High Dimensional Probability and Statistics, Average-Case Complexity, Statistical Physics

Education

Sep 2024–Present **Stanford University**
PhD in Computer Science, Theory Group
Advisors: Tselil Schramm & Nima Anari
GPA: 4.0/4.0

Jul 2017–May 2022 **Indian Institute of Technology Kanpur**
BTech in Electrical Engineering & BS in Mathematics
8 Semesters; Overall GPA: 9.3/10; Math GPA: 9.8/10

Jan 2020–Dec 2020 **Aalto University**
Academic Exchange, School of Science
2 Semesters; GPA: 4.78/5

Experience

Jul 2022–Aug 2024 **Google DeepMind, India**
Pre-Doctoral Researcher, Machine Learning and Optimization (MLO) Team

Jul 2021–Dec 2021 **Max Planck Institute for Intelligent Systems, Germany**
Internship; Hosts: Bernhard Schölkopf & Michael Muehlebach

Apr 2021–Jun 2021 **Tata Institute of Fundamental Research, Mumbai**
Internship; Host: Sandeep Juneja

Publications and Preprints

[1] **Averaged Annealed Markov Chains Achieve the Algorithmic Threshold for Tensor PCA, Part II: Langevin Dynamics**
with Tselil Schramm
Under Review · 2026 Paper ↗

[2] **Averaged Annealed Markov Chains Achieve the Algorithmic Threshold for Tensor PCA, Part I: Glauber Dynamics**
with Tselil Schramm
Under Review · 2026 Paper ↗

[3] **Parallel Sampling from the Ising p-Spin Model**
with Nima Anari and Alireza Haqi
International Conference on Randomization and Computation (**RANDOM**) 2026 Paper ↗

[4] **Diffusion Models are Secretly Exchangeable: Parallelizing DDPMs via Auto Speculation**
with Hengyuan Hu, Dorsa Sadigh, and Nima Anari
International Conference on Machine Learning (**ICML**) 2025 Paper ↗

[5] **Near-Optimal Streaming Heavy-Tailed Statistical Estimation with Clipped SGD**
with Dheeraj Nagaraj, Soumyabrata Pal, Arun Sai Suggala, and Prateek Varshney
Neural Information Processing Systems (**NeurIPS**) 2024 Paper ↗

[6] **Provably Fast Finite Particle Variants of SVGD via Virtual Particle Stochastic Approximation**
with Dheeraj Nagaraj
Neural Information Processing Systems (**NeurIPS**) 2023 · **Spotlight** Paper ↗
Oral at Optimal Transport and Machine Learning Workshop, NeurIPS 2023

[7] **Utilising the CLT Structure in Stochastic Gradient based Sampling : Improved Analysis and Faster Algorithms**
with Dheeraj Nagaraj and Anant Raj
Conference on Learning Theory (**COLT**) 2023 Paper ↗

[8] **Near Optimal Heteroscedastic Regression with Symbiotic Learning**
with Dheeraj Baby, Dheeraj Nagaraj, and Praneeth Netrapalli
Conference on Learning Theory (**COLT**) 2023 Paper ↗

[9] **Sampling without Replacement Leads to Faster Rates in Finite-Sum Minimax Optimization**
with Bernhard Schölkopf and Michael Muehlebach
Neural Information Processing Systems (**NeurIPS**) 2022 Paper ↗

[10] **NeurInt - Learning Interpolation by Neural ODEs**
with Avinandan Bose, Yatin Dandi, and Piyush Rai
NeurIPS 2021 DL & Differential Equations Workshop · **Spotlight** Paper ↗

[11] **TorchGAN: A Flexible Framework for GAN Training and Evaluation**
with Avik Pal
Journal of Open Source Software (**JOSS**) 2021 Paper ↗

[12] **Jointly Trained Image and Video Generation using Residual Vectors**
with Yatin Dandi, Soumye Singhal, Vinay P. Namboodiri, and Piyush Rai
Winter Conference on Applications of Computer Vision (**WACV**) 2020 Paper ↗

Talks

Nov 2026 **Reaching the Algorithmic Threshold for Tensor PCA via Local Markov Chains**
Stanford Theory Seminar; UC San Diego Theory Seminar

Aug 2024 **Linear Time Streaming Algorithms for Heavy Tailed Statistics**
MSR-IISc Theory Seminar Slides ↗ · Video ↗

Nov 2023, Mar 2024 **Sampling Through the Lens of Optimization: Recent Advances and Insights**
MSR-IISc Theory Seminar; EPFL FLAIR Seminar Slides ↗ · Video ↗

Jul 2023 **Utilising the CLT Structure in Stochastic Gradient-Based Sampling**
Conference on Learning Theory Slides ↗

Jul 2023 **Near-Optimal Heteroscedastic Regression with Symbiotic Learning**
Conference on Learning Theory Slides ↗

Service and Mentoring

CS Graduate Mentor · Stanford University
Reviewer · NeurIPS 2023, AISTATS 2022, NeurIPS 2021
Coordinator · Special Interest Group in Machine Learning, IIT Kanpur
Project Mentor · Programming Club and Math Club, IIT Kanpur

Selected Honors

2024 **School of Engineering Fellowship** · Stanford University
2018–20 **Academic Excellence Award** · IIT Kanpur
2016 **KVPY Scholarship** · Government of India